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Goals Report

In Partnership with ElectroRoute



ElectroRoute

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4E4 Industrial Internship Project

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1 Company Background

1.1 Company Introduction

ElectroRoute Holdings Limited is an international energy trading and services company owned by the Mitsubishi Corporation. The company mission is "to solve the commercial mechanics of a decarbonised energy system". By combining deep expertise in energy markets with cutting-edge digital infrastructure, ElectroRoute helps their clients maximise the value of their renewable assets, navigate market volatility, and contribute to system-wide decarbonisation.

The company operates both proprietary trading and physical shift-based trading functions. Proprietary trading involves trading the firm's own capital to take positions in energy markets based on price signals, forecasts, and market fundamentals, with the objective of generating profit. Shift trading primarily relates to the physical trading and real-time management of electricity and gas positions. This involves the continuous, 24/7/365, execution of trades to balance physical supply and demand, manage forecast deviations, and respond to real-time system conditions. Shift traders are responsible for ensuring that physical delivery obligations are met, while managing imbalance risk and maintaining compliance with market rules.

As at March 2025, the ElectroRoute had a turnover of €753.7 million and 124 employees. The company has offices in Dublin (headquarters), Letterkenny, London, and Tokyo. For the 2024/25 financial year, ElectroRoute managed over 2 Gigawatts of client assets and executed on average 5,200 trades per day. ElectroRoute currently holds a 3.5 GW portfolio of physical assets, and it is set to grow further. The company currently operates 79 MW of renewables in the GB market, which is set to grow to 531 MW by FY26/27.



Figure 1: Current overview of ElectroRoute's assets and some of their clients.

1.2 Company Services

ElectroRoute offers a comprehensive range of energy trading and market services designed to support the transition to a more sustainable energy system. These include:

Power, green, and gas trading:

- Trading of electricity, renewable certificates, and gas across wholesale markets to manage price risk and optimise revenues.
- Participation in day-ahead, intraday, and forward markets using market analytics and forecasting.
- Management of renewable energy guarantees to support low-carbon market integration.

Corporate Power Purchase Agreements (PPAs):

- Structuring long-term electricity supply agreements between renewable generators and corporate off-takers.
- Acting as an intermediary to balance risk, manage market access, and handle settlement for both parties.
- Providing revenue certainty for renewable assets while supporting corporate sustainability objectives.

Fixed price revenue guarantee structures:

- Contractual structures that provide renewable generators with a guaranteed minimum revenue over a defined period.
- ElectroRoute assumes wholesale price risk in exchange for potential market upside.
- Improving project financial viability and supporting renewable asset financing.

Renewable balancing and market access:

- Enabling renewable generators to participate in electricity markets despite variable and uncertain output.
- Provision of forecasting, nomination, and imbalance management services.
- Management of imbalance risk to support system stability and regulatory compliance.

Battery and flexible asset optimisation:

- Optimisation of batteries and flexible assets across multiple revenue streams.
- Trading of these assets in energy, balancing, and ancillary service markets.
- Use of optimisation algorithms to maximise asset value while supporting grid flexibility.

2 Internship Project Area Background

My internship is based within ElectroRoute's trading operations and market analytics functions, which sit at the core of the company's commercial activity. These teams are responsible for the real-time and forward management of physical energy positions across electricity and gas markets, as well as the analysis and optimisation of renewable and flexible assets. The role operates at the intersection of energy systems, financial markets, and data-driven decision-making.

A key focus of the project area is physical trading and asset optimisation. This involves forecasting renewable generation and demand, monitoring system conditions, managing imbalances, and executing trades to ensure that physical delivery obligations are met in real time. Due to the variable nature of renewable generation, effective forecasting, monitoring, and rapid response to market conditions are critical to minimising imbalance costs and maximising asset value.

The internship also sits heavily within the domain of data analysis and automation. ElectroRoute maintains large databases populated with live and historical market, weather, and asset data. These datasets are used to produce dashboards, key performance indicators (KPIs), trading lookaheads, and analytical tools that support both shift traders and proprietary trading teams. As a result, strong technical skills in Python, SQL, and Excel are essential within the project area.

Through this internship, I am embedded in an environment that combines operational responsibility with analytical problem-solving. The project area provides exposure to real-world energy markets, the challenges associated with operating a decarbonised energy system, and the practical application of engineering, statistical, and computational techniques to commercial decision-making.

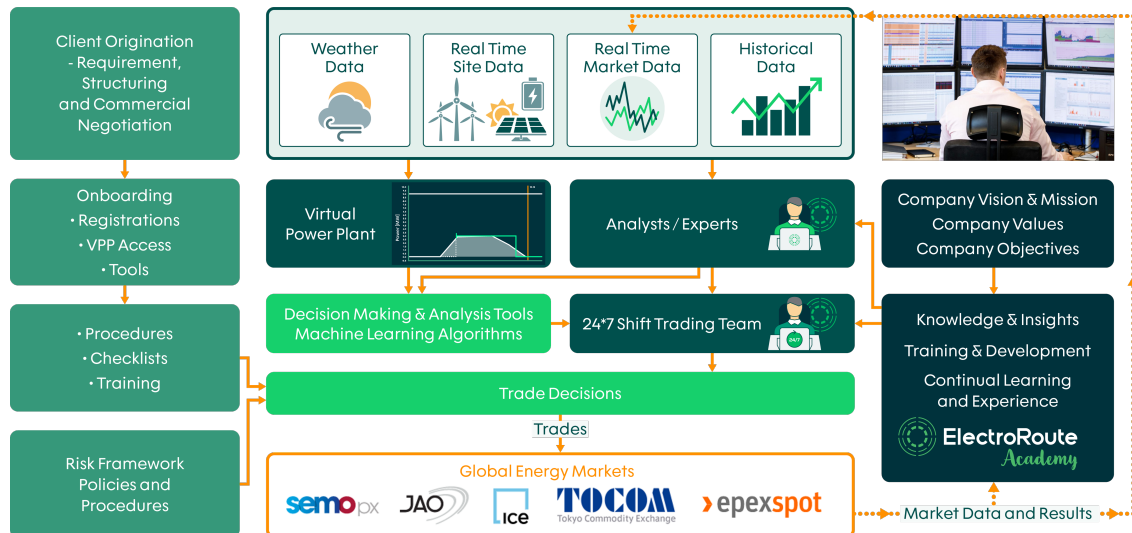


Figure 2: Flow chart describing the processes ElectroRoute uses to make decisions on the trading floor.

3 Project Details

3.1 Aims & Objectives

Below are some of the goals I aim to achieve during my time at ElectroRoute.

Complete the ElectroRoute Academy Trader Training:

- The ElectroRoute Academy is the company’s in-house trader training programme designed to develop traders from entry level to full operational capability.
- The programme typically spans a 12-month period and is divided into pre go-live and post go-live training phases on the operational trading desk.
- During the pre go-live phase (approximately 1–3 months), trainees focus on building a strong foundation in energy markets, power systems, weather systems, ElectroRoute’s internal systems and tools, risk and compliance, probability and statistics, and the operational aspects of the trading role.
- This phase also places significant emphasis on developing technical skills in technologies such as Python, SQL, VBA, and Excel.
- During the post go-live phase (approximately 3–12 months), traders begin engaging with live operational trading activities and undertake a wide range of projects, including the analysis of extreme market events and the development of Python- and Excel-based trading tools.
- In order to begin trading independently, traders must become certified by passing the external EPEX SPOT examination.
- While I will not be executing trades as an intern, I aim to work towards passing the EPEX SPOT exam as a measure of the technical and market knowledge gained during my internship.

Develop my Technical Skills:

- A significant component of trading and market analysis involves the development of strong technical and programming skills.
- ElectroRoute makes extensive use of technologies such as Python, SQL, VBA, and Excel to analyse and manipulate large datasets, develop dashboards, and support trading strategies and algorithms.
- As part of my internship, I have been tasked with completing a Dataquest course titled *Data Scientist in Python*, which covers topics ranging from data analysis using libraries such as NumPy and Pandas to the development of statistical machine learning algorithms.
- ElectroRoute maintains large databases populated with live market and asset data, and I have been developing skills in querying these databases using SQL and importing the data into Python to build analysis tools and dashboards.

Produce Valuable Contributions to the Trading Team:

- Building on the above points, I aim to leverage what I have learned during my training and produce valuable work that can aid the team's operation.
- This can range from building analysis tools, producing reports, delivering information to the team, and other activities outlined in sections 3.2 and 3.4.

3.2 Tasks

Below are some of the tasks that I am responsible for completing during my time at ElectroRoute, outlined by my manager. The list is subject to change, however it provides a good basis overview.

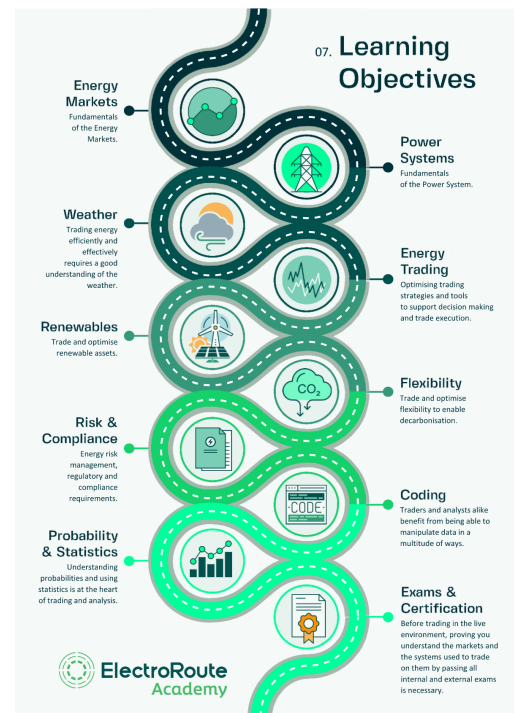


Figure 3: Trader training pathway

Role	Trading Operations Intern
Job Brief	– Support the daily asset trading activity by performing daily and weekly checks, conducting analysis, and reporting metrics to the wider team.
Daily Tasks	– Perform daily market analysis on Irish and GB power markets, as directed by the shift trader or shift trading manager, to support decision-making and daily trading strategies. – Confirm SCADA availability for all units and follow up with clients where data is unavailable. – Check all units for outages and follow up with clients where discrepancies are identified.
Weekly Tasks	– Compile and present team KPIs at the weekly meeting. – Prepare and present weekday (Monday) and weekend (Friday) trading lookaheads. – Review and summarise SEMO constraint updates, ESB outage reports affecting assets, REMIT outage reports, and BSRM interconnector outage reports. – Review industry news, articles, and blogs, and share key insights via an internal weekly email.
Other Tasks	– Support analysis projects as directed by the team. – Assist with trading KPI reporting and performance tracking, including script development and handover from the previous Renewables Manager. – Attend SEMO market calls and report relevant updates (e.g. MOUG, Market Participant Calls, and Future Markets Updates). – Build and maintain analysis tools in Python and/or Excel.

Table 1: Summary of Trading Operations Intern Role

Some other tasks that were outlined in my induction brief included:

- Completing the new user self setup. This is a comprehensive guide written to help new employees get set up technologically. This includes setting up tools such as Python and SQL, and gaining access to various software and databases used by ElectroRoute.
- Getting started with Dataquest's Python for data science course.
- Beginning the EFET (European Federation of Energy Traders) Energy Markets Training Programme. This course is a guide to understanding commodity markets.
- There are also other trainings within the academy such as ISEM training, weather training, etc..
- Attending meetings with IT and compliance.
- Getting started with my own paper trading project once set up and up to speed.
- Shadowing one of the traders on desk for a few hours to understand what it is they do.
- Learning how to access the VPP (virtual power plant) and how to read SCADA and report outages.
- Learning how to produce graphs using the database and Python to present KPIs.
- Learning how to analyse the market's future conditions, produce look-aheads, and present findings to the team.
- Attending any industry seminars and calls such as SEMO market calls to stay up to date with current news and regulation.

3.3 Deliverables

The following deliverables are expected to be produced over the course of the internship. These deliverables reflect both tangible technical outputs and demonstrated competency in trading operations and analysis.

- Completion of the pre go-live components of the ElectroRoute Academy trader training, including core market, systems, and compliance modules.
- Completion of assigned technical training courses, including Dataquest’s *Data Scientist in Python*, with evidence of applied learning.
- Development and maintenance of Python- and/or Excel-based analysis tools to support trading operations, KPI reporting, or market analysis.
- Regular production of market analysis outputs, including daily market summaries, trading lookaheads, and KPI reports.
- Contribution to team reporting through the preparation and presentation of weekly KPIs and trading outlooks.
- Demonstrated ability to query company databases using SQL and integrate data into analytical workflows.
- Documentation of tools, scripts, or analytical methods developed during the internship to support knowledge transfer within the team.
- Active participation in market calls, meetings, and operational processes, with relevant insights communicated to the trading team.

3.4 Project Timeline

The internship broadly aligns with the structure of the ElectroRoute Academy trader training programme. The timeline is divided into an initial training-focused phase followed by a longer period of applied trading analysis and project-based work.

Months 1–3: Foundation and Pre Go-Live Training

The initial phase of the internship focuses on building a strong foundation in energy markets and trading operations. This includes completion of the pre go-live components of the ElectroRoute Academy, covering electricity market structures, power systems, weather and forecasting, risk and compliance, and the operational responsibilities of the trading desk. During this period, I will also develop core technical skills in Python, SQL, VBA, and Excel, with an emphasis on analysing market data, automating workflows, and understanding existing trading tools and dashboards. Shadowing shift traders will provide exposure to live market operations and decision-making processes.

Month 3 Onward: Applied Trading, Analysis, and Project Work

The second phase of the internship is focused on applying the acquired knowledge in a practical trading and analysis context. This includes the development and enhancement of Python- and Excel-based trading and analysis tools, supporting daily and weekly trading operations, and contributing to KPI reporting and performance tracking. Project-based work during this phase may include paper trading exercises, analysis of historical and extreme market events, and the optimisation of trading strategies using historical data. I will also continue to engage with live operational processes, market calls, and trading lookaheads, with increasing responsibility and independence over time.